```
      .attr("height", thisBounds[1][1] - thisBounds[0][1])
         .style("fill", "none")
         .style("stroke-dasharray", "5 5")
         .style("stroke", "black")
         .style("stroke-width", 2)
      .style("pointer-events", "none");

  d3.select("svg")
    .append("circle")
    .attr("class", "centroid")
    .style("fill", "red")
    .attr("r", 5)
    .attr("cx", thisCenter[0]).attr("cy", thisCenter[1])
    .style("pointer-events", "none");
};
function clearCenterBounds() {
  d3.selectAll("circle.centroid").remove();
  d3.selectAll("rect.bbox").remove();
};
```

Centroid is an array with the x and y coordinates of the center of a feature

Removes the shapes when you mouse off a feature

You've learned the core geo functions that allow you to make maps with D3: `geo.projection` and `geo.path`. By using these functions, you can create maps with a distinct look and feel, and provide your users with the ability to interact with them as shapes and as geographic features. D3 provides more functionality, and we'll dive into it now.

7.2 Better mapping

To make your maps more readable, you can use built-in features from `d3.geo`: the graticule generator and the zoom behavior. One provides grid lines that make it easier

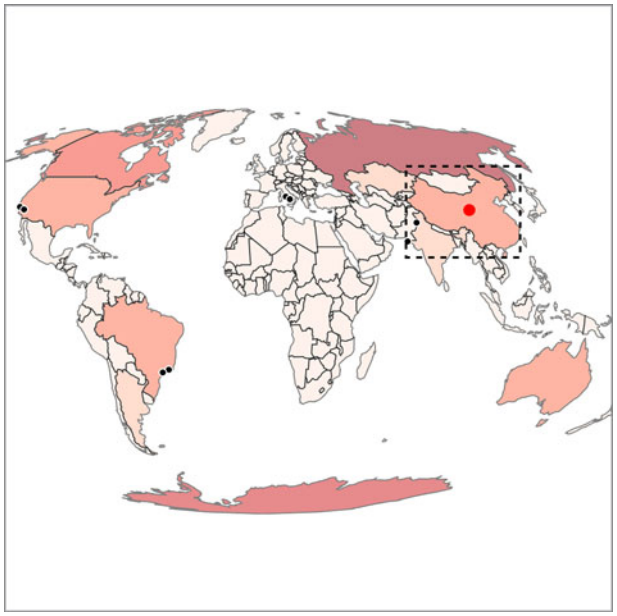


Figure 7.7 Your interactivity provides a bounding box around each country and a red circle representing its graphical center. Here you see the bounding box and centroid of China. The D3 implementation of a centroid is weighted, so that it's the center of most area, and not just the center of the bounding box.